$= 52 + (0.75)\ (57 - 52)$

$= 52 + 3.75$

$Q_3 = 55.75$

Quartiles for Discrete Series (grouped data)

Step 1 : Find cumulative frequencies

Step 2 : Find $\left(\frac{N+1}{4}\right)$

Step 3 : See in the cumulative frequencies, the value just greater than $\left(\frac{N+1}{4}\right)$, the corresponding value of x is Q_1

Step 4 : Find $3\left(\frac{N+1}{4}\right)$

Step 5 : See in the cumulative frequencies, the value just greater than $3\left(\frac{N+1}{4}\right)$ then the corresponding value of x is $Q3$.

Example 5.31

Compute Q_1 and Q_3 for the data relating to age in years of 543 members in a village

Age in years	20	30	40	50	60	70	80
No. of members	3	61	132	153	140	51	3

Solution:

x	f	cf
20	3	3
30	61	64
40	132	196
50	153	349
60	140	489
70	51	540
80	3	543

$Q_1 = \left(\frac{N+1}{4}\right)^{th}$ item

$= \left(\frac{543+1}{4}\right)^{th}$ item

$= 136^{th}$ item

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